

Full length article

Dietary supplementation of probiotics fermented Chinese herbal medicine *Sanguisorba officinalis* cultures enhanced immune response and disease resistance of crucian carp (*Carassius auratus*) against *Aeromonas hydrophila*

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ABSTRACT

Aeromonas hydrophila, a Gram-negative bacterium, is one of the major pathogens causing bacterial sepsis in aquatic animals due to drug resistance and pathogenicity, which could cause high mortality and serious economic losses to the aquaculture. *Sanguisorba officinalis* (called DiYu in Chinese, DY) is well known as herbal medicine, which could inhibit the growth of pathogenic bacteria, hemostasis and regulate the immune response. Moreover, the active ingredients in DY could remarkably reduce drug resistance. In this study, we investigated the effects of probiotic fermentation cultures on *A. hydrophila* through in vitro and in vivo experiments. Three lactic acid bacteria, including *Lactobacillus rhamnosus* (LGG), *Lactobacillus casei* (LC) and *Lactobacillus plantarum* (LP), were selected to ferment the Chinese herbal medicine DY. The assays of antagonism showed that all three fermented cultures could influence the ability of *A. hydrophila* growth, among which *L. rhamnosus* fermented DY cultures appeared to be the strongest inhibitory effect. In addition, the biofilm determination revealed that *L. rhamnosus* fermented DY cultures could significantly inhibit the biofilm formation of *A. hydrophila* compared to the other groups. Furthermore, protease, lecithinase and urease activities were found in the three fermentation cultures. Three probiotics fermented DY cultures were orally administration with crucian carp to evaluate the growth performance, immunological parameters and pathogen resistance. The results showed that the three fermentation cultures could promote the growth performance of crucian carp, and the immunoglobulins, antioxidant-related enzymes and immune-related genes were significantly enhanced. Besides, the results showed that crucian carp received *L. rhamnosus* (60.87%), *L. casei* (56.09%) and *L. plantarum* (41.46%) fermented DY cultures had higher survival rates compared with the control group after infection with *A. hydrophila*. Meanwhile, the pathological tissue results revealed that the probiotic fermented cultures could largely improve the tissues damage caused by the pathogenic bacteria. In conclusion, this study proved that the fermentation cultures of three probiotics could effectively inhibit the growth of *A. hydrophila*, regulate the level of immune response and improve the survival rate against *A. hydrophila* in crucian carp. The present data suggest that probiotic fermented *Sanguisorba officinalis* act as a potential gut-targeted therapy regimens to protecting fish from pathogenic bacteria infection.

1. Introduction

A. hydrophila is a common pathogen in freshwater environments [1],

which could cause aquatic animals infection, such as fish, shrimp and shellfish through adhesion [2], invasion, and toxin production of the pathogen [3]. Also, hemorrhagic septicemia [4], bacterial enteritis [5],

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and other diseases caused by the pathogenic *A. hydrophila* were frequently reported in mammals [6]. The applications of inactivated vaccines, attenuated live vaccines, subunit vaccines and antibiotics are common preventive and control methods used in aquaculture. However, some disadvantages such as short immunization time, virulence enhancement and poor immunogenicity, should not be ignored. Currently, numerous studies have demonstrated that long-period use of antibiotics leads to the development of multi-drug resistant bacteria [7]. Chinese medicine is considered an effective alternative antimicrobial agent, which could maintain immunological homeostasis and prevent against bacterial infection, colonization and inflammation [8]. Previous studies have shown that the potential of *S. officinalis* to substantially improve symptoms in ulcers, oxidative stress, diarrhea and bacterial infection [9,10]. As commensal bacteria in the gastrointestinal tract, some probiotics have immunomodulatory and intestinal flora maintenance characteristics, which could be considered as a fermentation solution for Chinese herbs [11]. Previous studies have confirmed the ability of probiotic fermented herbal cultures resistance against pathogens and enhance immunity [12]. In addition, probiotic fermented herbal complexes could regulate the balance of animal intestinal flora and promote the absorption and metabolism of drug components in the gastrointestinal tract [13]. Probiotic fermented herbal medicines could transform the large molecules, polysaccharides and proteins of herbal medicines, into small molecules that could be directly absorbed by the intestinal flora [14]. It is reported that fermentation of the herb *Paeoniae Radix* (PR) by *L. plantarum* 174A could regulate the expression of inflammatory cytokines and reduce oxidative damage of macrophages [15]. There is evidence that probiotics fermented herbal medicine plays a crucial role in regulating growth performance, serum parameters, immune-related cytokines and promotes intestinal health in broilers [16]. However, few researches have been able to draw on about probiotics fermented Chinese medicine against bacterial diseases in fish.

Here, we speculated that probiotic fermented Chinese herbal could enhance the immune responses and resistance against *A. hydrophila*. To further analyze this hypothesis, three potential probiotic fermented cultures were administration with fish. The effects of probiotic fermentation on growth performance, cytokines gene expression and disease resistance in Crucian carp were investigated. Therefore, the present study contributes to provide evidence that immune regulation and protective effect in fish mediated by probiotic fermented Chinese herbal product that enhances innate immune response and prevents pathogen infection.

2. Materials and methods

2.1. Bacterial strains and growth conditions

L. rhamnosus ATCC 53103 (LGG) and *L. casei* ATCC 393 (LC) were purchased from the China General Microbiological Culture Collection Center. *L. plantarum* H8 (LP) and *A. hydrophila* TPS were provided by the Laboratory of Preventive Veterinary Medicine, Jilin Agricultural University. *L. rhamnosus*, *L. casei* and *L. plantarum* were cultured anaerobically without shaking in De Man, Rogosa and Sharpe (MRS) medium (Solarbio, China) at 37 °C. *A. hydrophila* TPS was grown in Luria-Bertani (LB) medium (Tiangen, China) at 30 °C.

2.2. Preparation of fermentation cultures and diets

The three lactic acid bacteria were diluted at 10^{-5} , 10^{-6} , and 10^{-7} CFU/mL using PBS (Phosphate buffer saline) buffer, respectively. The diluted suspension was spread on MRS agar, anaerobically incubated at 37 °C for 24 h and colonies were counted. *L. rhamnosus* (6×10^8 CFU/mL), *L. casei* (6.2×10^8 CFU/mL) and *L. plantarum* (6×10^9 CFU/mL) were adjusted to 6×10^8 CFU/mL. The herb DY purchased from Jilin Pharmacy, was dried and particle through an 80-mesh sieve. Based on the preliminary experiments, the optimal fermentation conditions were

as follows: the inoculum of probiotics was 4% and the content of herbal medicine was 8%. The 8 mL of lactic acid bacteria and 16 g of herbal powder were anaerobically incubated at 37 °C for 24 h. The obtained fermentation cultures were centrifuged at 5500 rpm (4 °C, 10 min) to separate supernatant. The prepared fermentation solution mixed with the basal diet to prepare the fish diet for the fermentation groups.

2.3. Fish and ethics statement

In the experiment, a total of 240 crucian carp, weighing 56 ± 5.0 g and 15.5 ± 1.2 cm in length were provided by a commercial farm in Jilin Province, China. These fish were kept randomly in four tanks (90 cm in diameter), which were equipped with a circulation system. The temperature of the water was controlled at 27 ± 1.0 °C and the pH value was maintained at 7.2 ± 0.5 . Fish adapted to this condition for two weeks. The fish in the control group were fed with the basal diet (purchased from Tongwei Co., Ltd., Changchun, China) during the immunization period (Table 1). Fish in the fermentation group were fed a basal diet mixed with fermented DY cultures (Table 2). After two weeks, the fish were received with the basal diet same as the control group. The crucian carp were fed approximately 3–4% of body weight twice a day (06:30 and 18:30). The animal experiment was performed in accordance with the Regulations for Animal Experimentation of Jilin Agriculture University (JLAU08201409).

2.4. Sampling and growth performance

Three fish from each group were randomly selected at 0 d, 14 d, 28 d, 42 d and 56 d after immunization, and anesthetized with 300 mg/L Methane-Sulfonate-222 (MS-222) and weighed. The liver, spleen, kidney, gut and gill tissues were collected and stored at -80 °C for further analysis. The collected blood was centrifuged at 3500 rpm for 10 min to obtain serum, and stored at -20 °C. The average weight gain rate (AWGR) and specific growth rate (SGR) were calculated using the following formulations.

$$AWGR = (W_t - W_0) / W_0 \times 100\%$$

$$SGR = (\ln W_t - \ln W_0) / t \times 100\%$$

W_0 and W_t represent the initial and final body weights of crucian carp (g/tail), and t represents the days of rearing.

2.5. Antimicrobial ability

The antagonism against the pathogen of fermented herbal medicine was performed as described previously [17]. Briefly, *A. hydrophila* was cultured in LB broth at 30 °C for 18 h. The cultured bacteria were diluted in PBS buffer to a final concentration of 10^8 CFU/mL, and 200 μ L bacterial suspension was spread on the agar plates. Then the equal volume of fermentation cultures was added to each well (3 mm in diameter) of agar plates, and PBS buffer was used as a negative control. After 18 h incubation at 30 °C, the inhibition zone diameter was measured in millimeters. The determination was repeated three times under the same conditions.

Table 1
Basic diet composition.

Ingredients	Nutritional Composition	Percentage of nutrients (%)
soybean meal	crude protein	$\geq 30.0\%$
fish meal	crude fiber	$\leq 7.0\%$
wheat flour	crude ash	$\leq 15.0\%$
Canola meal	crude fat	$\geq 3.0\%$
yeast hydrolysate	moisture	$\geq 10.0\%$
sodium chloride	phosphorus	$\geq 1\%$
Vitamins	lysine	$\geq 1.5\%$
trace elements	calcium	0.5%

Table 2

Experimental treatment of different probiotic fermented herbal medicines DY as dietary supplements.

Treatments	Diets
Control	Basal feed
GD	Basal feed mixed with <i>L. rhamnosus</i> fermented DY cultures
CD	Basal feed mixed with <i>L. casei</i> fermented DY cultures
PD	Basal feed mixed with <i>L. plantarum</i> fermented DY cultures

2.6. Minimum inhibitory concentration (MIC) and minimum bactericidal concentration determination (MBC)

MIC and MBC of the fermented cultures were determined by the microdilution method, as described by previous studies [18,19]. Briefly, the fermented cultures were 10-fold serially diluted, and the concentration range of 1–0.0625 µg/mL, and dispensed in test tubes. Subsequently, 40 µL of *A. hydrophila* was added and incubated at 30 °C for 16–20 h. The highest dilution concentration without bacterial growth is regarded as the MIC. MBC was determined by colony growth. The liquid in tubes with concentrations more than MIC was spread onto LB agar and incubated overnight. The concentration of the first non-growing colony on MRS agar was MBC [20].

2.7. Analysis of enzyme activity assay

According to the previously reported method [21], the activities of lipase, protease, gelatinase, lecithinase, urease, and amylase in fermentation cultures were determined by the Oxford cup diffusion method. Briefly, LB agar containing soluble starch (2%), Tween 80 (1%), skim milk (8%), gelatin (4%), egg yolk (5%), and urea (2%) were prepared, respectively. In agar plates with 6 mm diameter wells, 100 µL of fermentation broth was added. Three lactic acid bacteria and DY aqueous decoction were used as blank control and incubated at 30 °C for 24–48 h.

2.8. Biofilm detection

To estimate the effect of different ratios of fermentation cultures on the biofilm formation of *A. hydrophila*, the biofilm assays were conducted using crystalline violet staining as reported by Zhang et al. [22]. Briefly, the ratio of the three lactic acid bacteria fermentation DY cultures and *A. hydrophila* was set up five gradients, 10 %–50%, and incubated overnight at 30 °C in LB medium. *A. hydrophila* was added to a 96-well polystyrene microplate and OD at 575 nm was measured using an enzyme mark instrument (Thermo Fisher Scientific) as a control. Afterward, *A. hydrophila* and fermentation cultures were added at equal ratio to a new 96-well polystyrene microtiter plate. The plates were washed three times with PBS buffer and fixed in methanol for 15 min. After the samples were air-dried, 200 µL 2% crystal violet was added to each well and stained for 20 min. The plate was washed with distilled water and 200 µL 33% glacial acetic acid was added to dissolve the crystal violet. Finally, the absorbance at 575 nm was measured using an enzymatic calibrator (Thermo Fisher Scientific). The detection was repeated three times.

Table 3

The primer sequence of qRT-PCR.

Gene	Sequence (5'-3') Forward	Sequence (5'-3') Reverse	GenBank No.
<i>MyD88</i>	GCCGACTCGCAAGTTGACTT	GCTTTTTCCTCAGCTCATGT	GR679416.1
<i>IL-10</i>	AACGTATGACCCGAATGGAAC	CACCTTCTCCAGTCGTCAA	HQ258106.1
<i>IL-1β</i>	CATCAAAGAAATCGCTCCTG	GCAAGGTCTGCCTGGTCT	AJ249137.1
<i>IFN-γ</i>	AACAGTCGGGTGTCGCAAG	TCAGCAAACATACTCCCCAG	EU069818.1
<i>TNF-α</i>	TTATGTCCGTGCGGCCCTTC	AGGTCTTCCGTGTGCGCTTT	EU909368.1
<i>β-actin</i>	CAAGATGATGGTGTGCCAAGTG	TCTGTCTCCGGCAGCAAGTA	AB039726.2

2.9. Non-specific immune response

Serum samples collected at 14 d, 28 d, 42 d and 56 d were used to detect Alkaline phosphatase (AKP), Superoxide dismutase (SOD), Catalase (CAT), Glutathione peroxidase (GSH-PX), and Lysozyme (LYS) activities and the levels of immunoglobulin M (IgM), complement C3 and C4. In the same method as previously reported [23], according to the instructions, ELISA kits (Nanjing Jiancheng Institute of Biological Engineering, Nanjing, China) were used for the assays. For all samples, repeat the experiment three times.

2.10. Immune-related gene expression

Total RNA was extracted from different tissues using the High Purity RNA Tissue Kit (Takara, Japan). The cDNA was synthesized using Nanodrop 2000c (Thermo Scientific, USA) to determine the concentration of RNA samples, according to the manufacturer's instructions [24], using the Reverse Transcriptase M-MLV kit (Takara, Japan). qPCR was performed using the 7500 system (Applied Biosystem, USA) for qPCR. The expression levels of *IL-1β*, *TNF-α*, *IFN-γ*, *IL-10* and *MyD88* genes were quantified. The β -actin gene was used as an internal control gene for qRT-PCR. GenBank registration information of the target genes and primer sequences are shown in Table 3. Each sample was repeated three times and quantified using the $2^{-\Delta\Delta Ct}$ method as described by Zhao et al. [25].

2.11. Challenge and histopathological analysis

After 56 days of immunization, to evaluate the protective effect of the three fermentation cultures, the crucian carp of all groups were intraperitoneally injection with pathogenic *A. hydrophila* TPS. After 14 days of observation, the survival rate of each group was measured. Referring to the method of Zhang et al. [26], the crucian carp fasted for 48–96 h before challenge. *A. hydrophila* TPS ($LD_{50} = 1 \times 10^5$ CFU/mL) was incubated overnight and adjusted to a concentration of 1×10^7 CFU/mL. Forty fish were randomly selected from each group and intraperitoneally injected with 100 µL bacterial suspension. The fish were injected with equivalent PBS buffer as the control group. Mortality was recorded and relative percentage survival (RPS, %) was calculated using the following formula. $RPS = (1 - \% \text{ mortality in immunization group} / \% \text{ mortality in control group}) \times 100$.

Meanwhile, liver, spleen, kidney and intestine tissues collected from each group of crucian carp after challenged were fixed in 4% neutral buffered formalin for 24 h and embedded in paraffin for histopathological analysis. Subsequently, the tissues were dehydrated in graded ethanol (70%, 80%, 85%, 90%, 95% and 100%) and washed with xylene as described using a tissue embedding machine (Leica Microsystems, Wetzlar, Germany) and prepared the serial sections (5 µm thick). Samples were stained with hematoxylin and eosin (H&E), for light microscopy analysis [27].

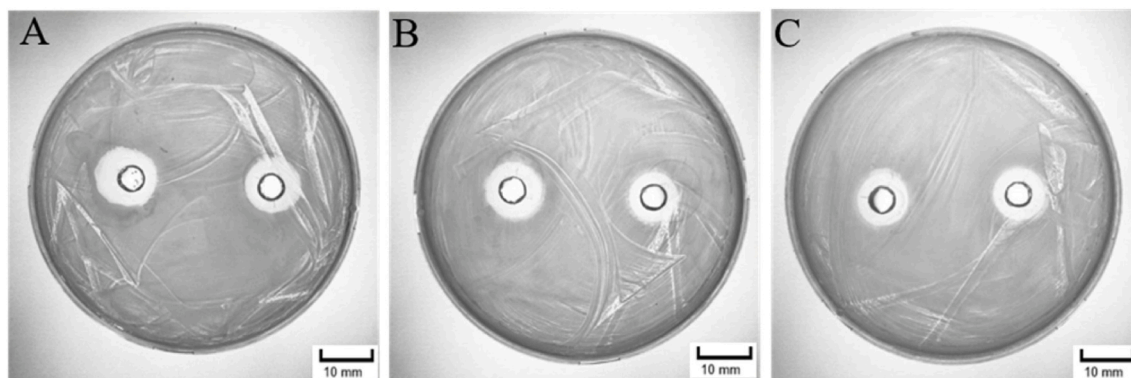
2.12. Statistical analysis

All statistical analyses were performed using GraphPad PRISM v8.0 and SPSS v.22.0 software. Variability was assessed by one-way analysis

Table 4

The effect of fermented Chinese medicine on the growth of crucian carp.

Index	Control	GD	PD	CD
Initial weight/g	54.59 ± 1.25 ^a	54.75 ± 0.74 ^a	55.03 ± 0.81 ^a	55.60 ± 0.62 ^a
Final weight/g	80.29 ± 0.10 ^a	85.05 ± 1.82 ^b	85.30 ± 1.83 ^b	85.22 ± 0.95 ^b
AWGR/% day ⁻¹	47.07 ± 3.77 ^a	55.39 ± 5.22 ^b	54.47 ± 4.29 ^b	53.29 ± 1.29 ^b
SGR/%/	0.68 ± 0.04 ^a	0.79 ± 0.06 ^b	0.78 ± 0.05 ^b	0.76 ± 0.02 ^b

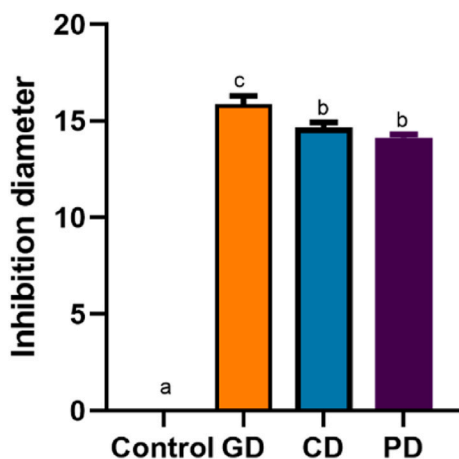
Different letters above represent significant differences between experimental groups ($p < 0.05$).**Fig. 1.** Inhibition of *L. rhamnosus* (A), *L. casei* (B), and *L. plantarum* (C) fermented herbal DY cultures against *A. hydrophila*.

of variance (ANOVA) and Tukey's test. Data are expressed as mean ± SD. $p < 0.05$ was considered a significant difference.

3. Result

3.1. Growth performance

The results of the three fermented cultures on the growth and rearing of crucian carp are shown in Table 4. After 56 days of administration, the AWGR and SGR of the GD, CD and PD groups were significantly higher than the control group ($p < 0.05$). There was no significant difference in AWGR and SGR among the three groups of fermented cultures, which further enriched the diversity of strain application for probiotics fermentation of herbal medicine.

**Fig. 2.** Inhibition zone diameters of *L. rhamnosus* (GD), *L. casei* (CD), and *L. plantarum* (PD) fermented herbal DY cultures against *A. hydrophila*. Data are expressed as mean ± standard deviation (SD) and results represent three independent experiments. Different letters above represent significant differences between experimental groups ($p < 0.05$).

3.2. Antimicrobial analysis

As shown in Fig. 1, the results showed that the *L. rhamnosus*, *L. casei*, and *L. plantarum* fermented herbal DY cultures all appeared significant inhibition with mean diameters of 15.85 mm, 14.56 mm and 14.13 mm, respectively. The results indicated that the fermented cultures could influence the growth of *A. hydrophila*. Among the three groups, the *L. rhamnosus* fermented cultures showed stronger antibacterial activity, which was significantly higher than the *L. casei*, and *L. plantarum* fermented cultures ($p < 0.005$). However, there was no significant difference between the *L. casei*, and *L. plantarum* fermented cultures (Fig. 2).

3.3. MIC and MBC

The MIC of *L. rhamnosus* fermented herbal DY cultures against *A. hydrophila* was 0.125 µg/mL, and the MIC of *L. casei* and *L. plantarum* fermented herbal DY cultures against *A. hydrophila* was 0.25 µg/mL (Table 5). The MBC of *L. rhamnosus* and *L. casei* fermented herbal DY

Table 5
MIC of fermentation cultures on *A. hydrophila*.

Group	Concentration	a	control
GD	1	–	+
	0.5	–	+
	0.25	–	+
	0.125	–	+
	0.00625	–	+
GD	1	–	+
	0.5	–	+
	0.25	–	+
	0.125	–	+
	0.00625	+	+
GD	1	–	+
	0.5	–	+
	0.25	–	+
	0.125	+	+
	0.00625	+	+

Positive (+): Turbidity indicating growth; Negative (–): No turbidity indicating absence of growth. a is the mean of three replicated experiments.

Table 6
MBC of fermentation cultures on *A. hydrophila*.

Group	Concentration	b	control
GD	1	–	+
	0.5	–	+
	0.25	–	+
	0.125	+	+
	0.00625	+	+
GD	1	–	+
	0.5	–	+
	0.25	–	+
	0.125	+	+
	0.00625	+	+
GD	1	–	+
	0.5	–	+
	0.25	+	+
	0.125	+	+
	0.00625	+	+

Positive (+): Indicating growth; Negative (–): Indicating absence of growth. b is the mean of three replicated experiments.

cultures against *A. hydrophila* was 0.25 µg/mL, and the MBC of *L. plantarum* fermented herbal DY cultures against *A. hydrophila* was 5 µg/mL (Table 6). Among the three groups of fermented cultures, the *L. rhamnose* fermented herbal DY showed stronger ability to inhibit *A. hydrophila* at lower concentrations.

3.4. Enzyme production

Plate-based activity assay was used to detect the enzyme activity of the fermentation cultures. The results suggested that *L. rhamnosus*, *L. casei* and *L. plantarum* fermented herbal DY cultures had strong protease (Fig. 3A, B and C), lecithinase (Fig. 3D, E and F) and urease (Fig. 3G, H and I) activities. However, there was no lipase, gelatinase and amylase activity observed in fermentation cultures (data not shown).

3.5. Biofilm assay results

The biofilm assay showed that different volume ratios of fermentation cultures could significantly affect the biofilm formation of *A. hydrophila*. Particularly, the ability of *A. hydrophila* biofilm formation

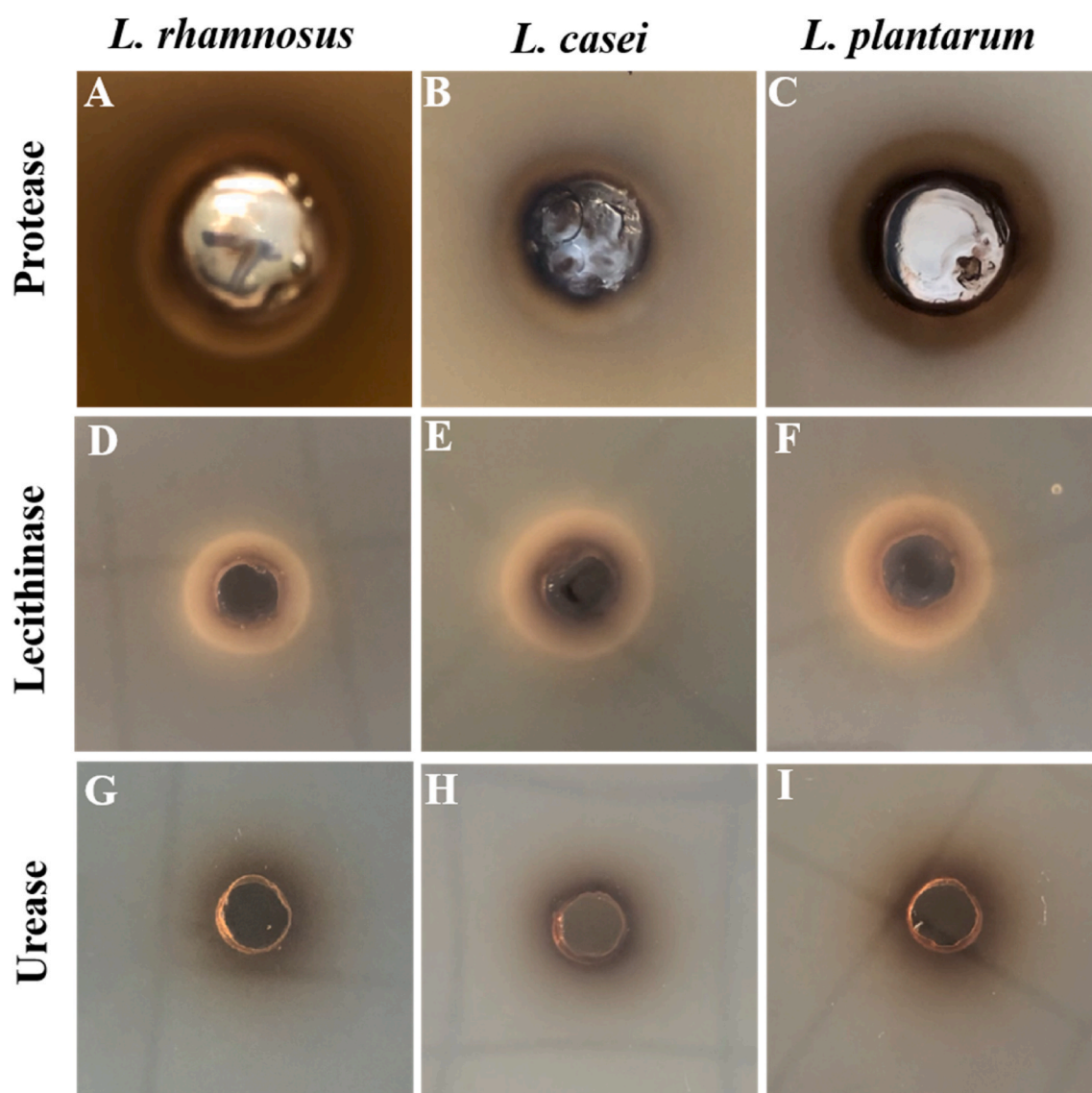


Fig. 3. Protease (A, B and C), lecithinase (D, E and F) and urease (G, H and I) assays results of *L. rhamnosus*, *L. casei*, and *L. plantarum* fermented herbal DY cultures.

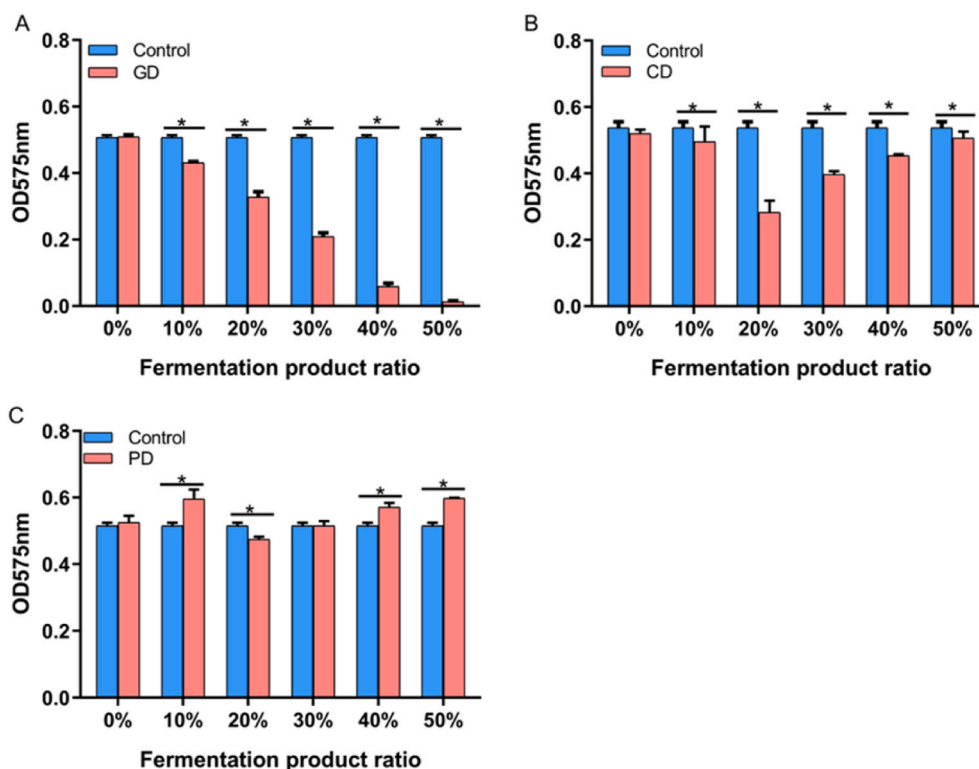


Fig. 4. Effect of *L. rhamnosus* (A), *L. casei* (B), and *L. plantarum* (C) fermentation of herbal DY cultures on the biofilm of *A. hydrophila*. * $p < 0.05$.

was significantly inhibited after treatment with DY cultures fermented by *L. rhamnosus* ($p < 0.05$) (Fig. 4A). The 20% ratio of *L. casei* fermentation DY cultures could effectively inhibit *A. hydrophila* biofilm formation (Fig. 4B). Nevertheless, the biofilm inhibition could be inhibited by high concentrations of probiotic fermented cultures. ($p < 0.05$) (Fig. 4B). We also noted that *L. plantarum* fermentation cultures and *A. hydrophila* volume at 20% effectively inhibited the biofilm formation of pathogenic bacteria, which was similar to the inhibitory effect of *L. casei* fermentation cultures ($p < 0.05$) (Fig. 4C). However, there was no significant inhibition at other proportions (Fig. 4C).

3.6. Non-specific immune analysis

To assess the immunogenicity of the probiotics fermented herbal DY cultures, which were used as a dietary supplement with crucian carp, and the serum non-specific immune parameters were determined using ELISA protocol. As illustrated in Fig. 5A, the serum AKP activity of crucian carp could be enhanced by probiotics fermented herbal DY cultures during immunization. At 56 days, the AKP activity of the GD, CD and PD groups was significantly higher than the control group ($p < 0.05$). And the AKP activity of the GD group was significantly higher than the CD and PD groups ($p < 0.05$). The SOD activity increased in GD, CD and PD groups after immunization and peaked at 56 days (Fig. 5B). Besides, the results revealed that the CAT activity of fish immunized with probiotics fermented cultures was significantly higher than control group before 14 days ($p < 0.05$) (Fig. 5C). In particular, the CAT activity of the GD group remained stable after 14 days, while the CAT activity of the PD group decreased after 28 days and stabilized after 42 days. It could be seen from the data in Fig. 5D that no significant difference of GSH-PX activity in the GD and CD groups was found compared with the control group before 14 days ($p > 0.05$). Further, higher level of GSH-PX activity were obtained on day 28 in the GD and CD groups after booster immunization compared to the control group ($p > 0.05$). The results of serum LYS levels are presented in Fig. 5E, the LYS levels of GD, CD and PD groups peaked at 42 days, which were significantly higher than the

control group ($p < 0.05$). However, no up-regulation of LYS level was found in the fermented cultures treated groups after 42 days post-administration. Oral administration of probiotics fermented herbal DY cultures could promote IgM secretion in the serum of crucian carp. At 56 days of immunization, the highest level of IgM was reached in the GD, CD and PD groups, which were significantly higher than the control group ($p < 0.05$) (Fig. 5F). The IgM levels in the GD group decreased after 28 days and re-increased after the booster immunization (day 42). There was a significant difference of C3 levels in the GD, CD and PD groups compared to the control group on day 42 ($p < 0.05$) (Fig. 5G). However, the C3 level of the GD group decreased after 42 days post-immunization. In Fig. 5H, there was a clear trend of increasing C4 levels in the serum of the fermentation group after 14 days of immunization. Notably, C4 levels were upregulated in the GD group, while decreased in the CD and PD groups after 28 days of immunization.

3.7. Cytokine expression analysis

Immune-related genes *IL-1 β* , *TNF- α* , *IFN- γ* , *IL-10* and *MyD88* expression in fish tissues were detected by RT-qPCR. After 14 days of immunization, the expression of *IL-1 β* in the liver, spleen, kidney and gills of GD, CD and PD groups were significantly higher than the control group ($p < 0.05$) (Fig. 6). *IL-1 β* expression of GD group was higher in the liver, kidney and gills compared to the CD and PD groups ($p < 0.05$). Further analysis showed that *IL-1 β* in kidney and gill tissues of the GD group was approximately twice as high as the CD and PD groups at 56 days (Fig. 6C and E). Also, *IL-1 β* expression in the PD group was approximately twice as high as the GD and PD groups in the gut at day 56 ($p < 0.05$) (Fig. 6D). From Figs. 7 and 8, the fermentation cultures up-regulated the expression of *TNF- α* and *IFN- γ* in different tissues, which was significantly higher than the control group on days 14, 28, 42 and 56 post-immunization ($p < 0.05$). And *TNF- α* and *IFN- γ* reached to peak at 56 days. The *IL-10* gene expression of the liver, spleen, kidney, gut and gills in the GD, CD and PD groups was significantly increased from the control group ($p < 0.05$) after 14 days of immunization (Fig. 9). Whereas

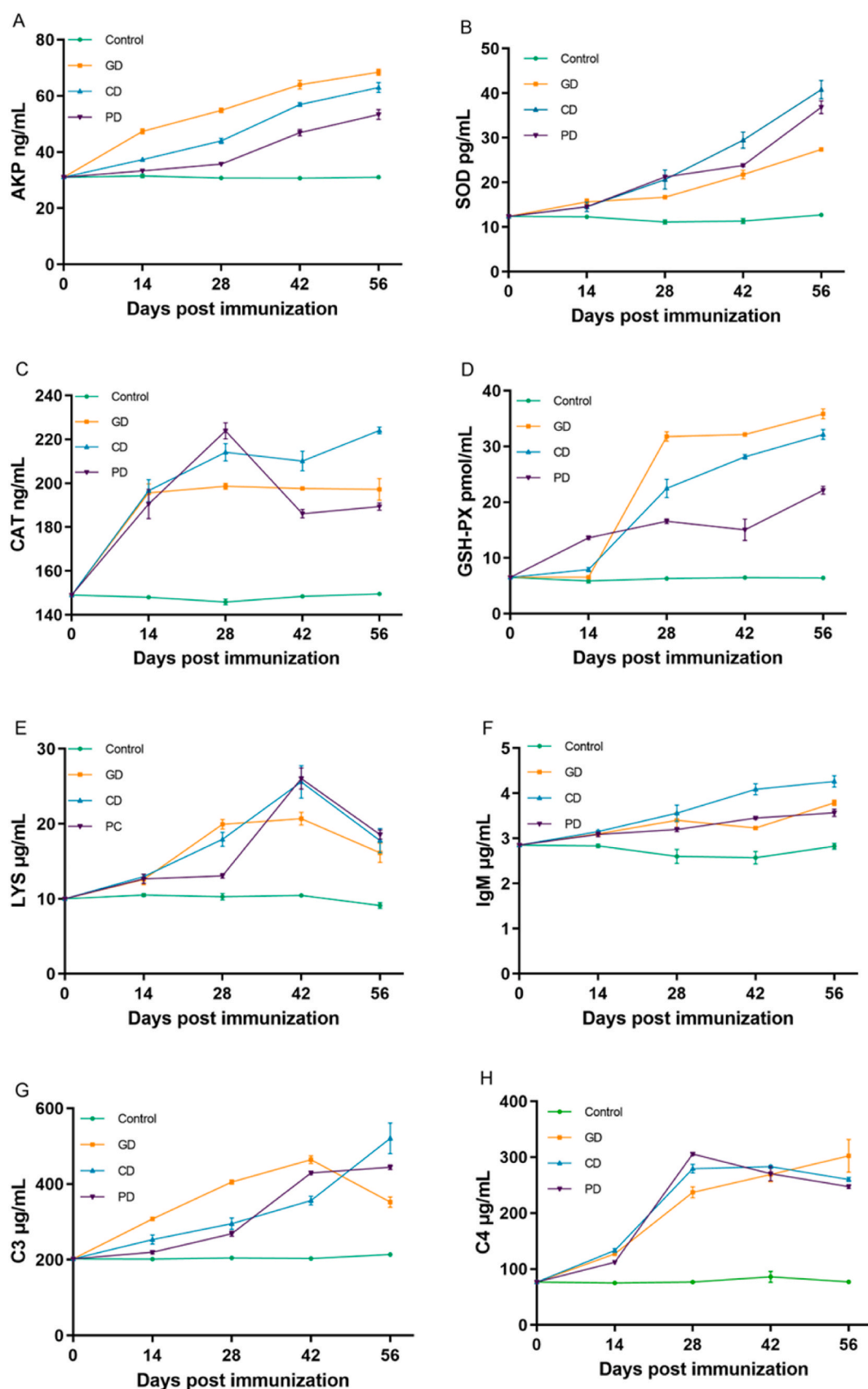


Fig. 5. Non-specific immune responses levels of AKP (A), SOD (B), CAT (C), GSH-PX (D), LYS (E), IgM (F), C3 (G) and C4 (H) stimulated by lactic acid bacteria fermented herbal DY cultures in crucian carp ($n = 3$ fish/group).

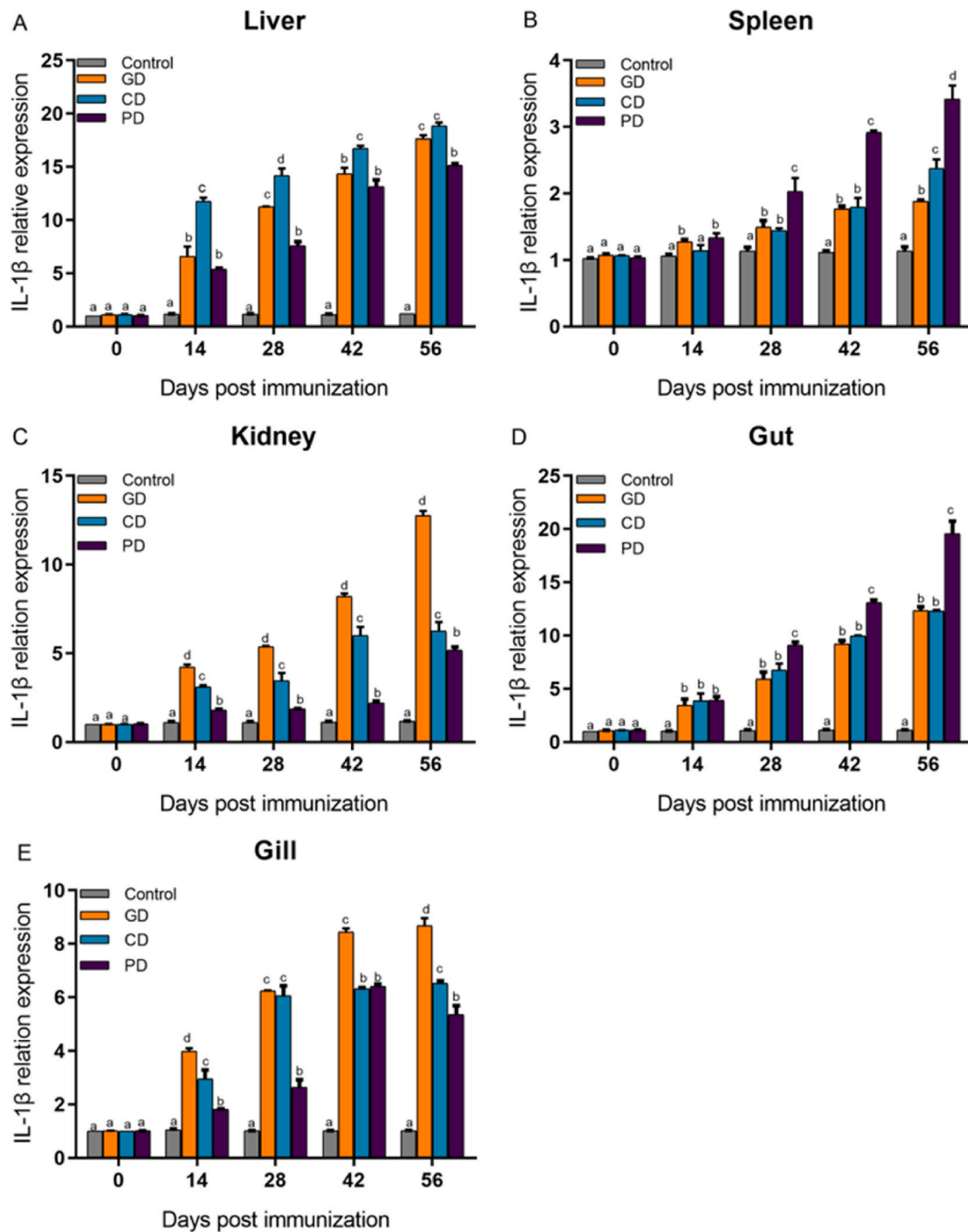


Fig. 6. qRT-PCR analysis of the expression of IL-1 β gene in Liver (A), Spleen (B), Kidney (C), Gut (D), and Gill (E) of crucian carp ($n = 3$ fish/group) after immunization. The values are expressed as the means $\pm$ SD ($n = 3$). Different letters placed above represent significant differences between the treatment groups ($p < 0.05$).

no significant differences were found among the GD, CD and PD groups (Fig. 9). *IL-10* gene expression increased exponentially in the liver, spleen, kidney and gills after 28 days of immunization compared to 14 days of immunization (Fig. 9A, B, C and E). Specifically, *IL-10* expression increased 1-fold in the liver and kidney and 2-fold in the spleen and gills (Fig. 9A, B, C and E). In Fig. 10, the results showed that the *MyD88* expression in the GD, CD and PD groups was significantly different from the control group ($p < 0.05$) in all tissues at 28 days. In addition, the GD group showed an increase in *MyD88* expression in the liver, spleen, intestine and gills during the immunization (Fig. 10A, B, D, and E), and the expression of the *MyD88* gene in the kidney was highest on day 28 (Fig. 10C).

3.8. Evaluation of protective efficacy

After 56 days of fermented cultures immunization with crucian carp, and fish were challenged with *A. hydrophila*. The survival rate of each group of crucian carp was observed after 14 days as shown in Fig. 11. The results displayed that all fish in the control group died within 7 days. Pathological sections could be seen that the fermented cultures relieved severe hemorrhage and tissue necrosis in the liver, spleen, kidney and gut compared to the control group (Fig. 12). We also found that the fermented culture was a good treatment for intestinal villi breakage, disorganization and intestinal bleeding in the gut (Fig. 12M, N, O and P). Suspension cultures were identified as *Aeromonas* spp., which was

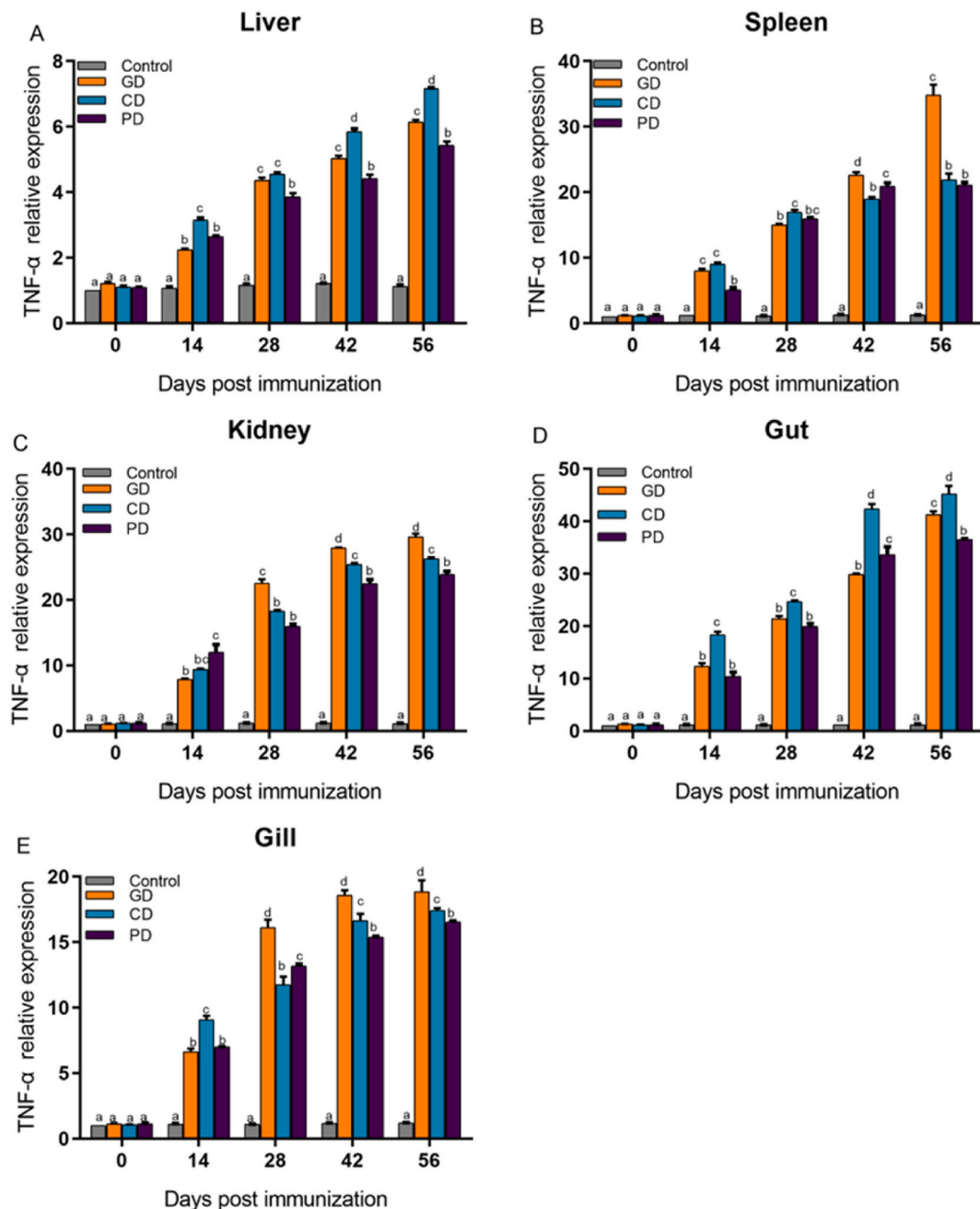


Fig. 7. qRT-PCR analysis of the expression of cytokine TNF- α gene in Liver (A), Spleen (B), Kidney (C), Gut (D), and Gill (E) of crucian carp ($n = 3$ fish/group) after immunization. The values are expressed as the means $\pm$ SD ($n = 3$). Different letters placed above represent significant differences between the treatment groups ($p < 0.05$).

confirmed by morphological observation and molecular identification of the colonies (data not shown). Mortality in the GD, CD and PD groups reached stability after 11 days post-infection. Further analysis revealed that the survival rates in the fish treated with fermented cultures groups (60.87%, 56.09%, 41.46%) were higher compared to the control group.

4. Discussion

Recently, antibiotics have been widely used in aquaculture. It is often used as a feed additive to prevent disease [28]. However, the retention and accumulation of antibiotics lead to pathogen resistance, immune disorders of aquatic animals, and threaten public health [29]. Probiotic fermented herbal medicine could promote the conversion of active substances in herbal medicine. Based on the degradation,

biofermentation, digestion and decomposition of microorganisms, the alkaloids, flavonoids, glycosides, amino acids, volatile oils and organic acids could be released from herbal medicines. Adding probiotic fermented herbal cultures to animal feed can improve animal immunity, prevent diseases and reduce feeding costs [30]. Previous studies have demonstrated that lactic acid bacteria fermented herbal medicines as diets could improve antioxidant capacity and immune response in broilers compared to separate fermented herbal medicine [31].

In this study, we confirmed that the fermentation of the herb *San-guisorba officinalis* (DY) by three probiotics as dietary supplements could improve the antioxidant capacity, immune response and disease resistance of crucian carp. Previously, chicks received with *Lactobacillus parviflorus* KL1 fermented herbal complex (ginseng, licorice, chickpea and scutellaria) could increase weight and specific growth rates [32]. In

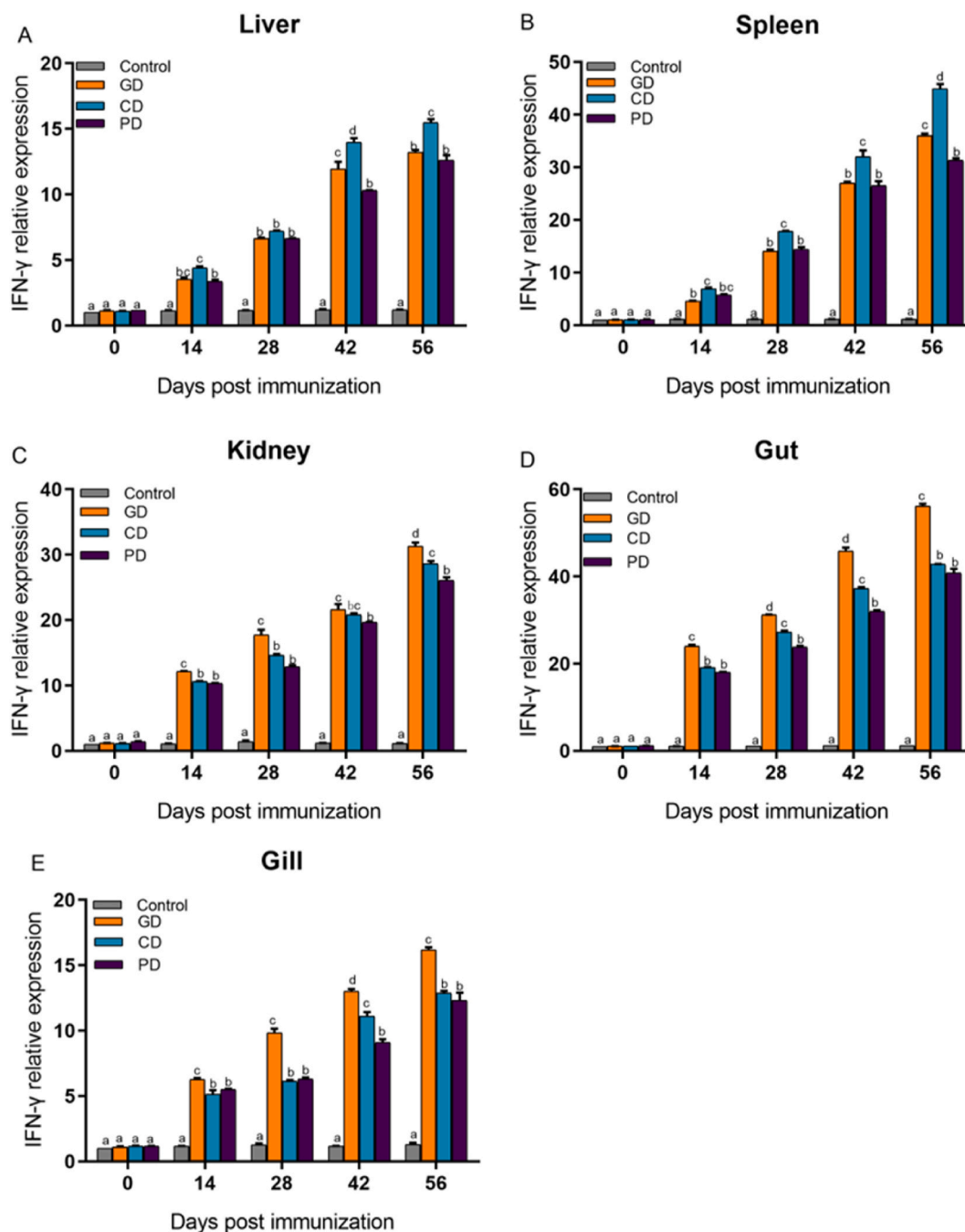


Fig. 8. qRT-PCR analysis of the expression of cytokine IFN- γ gene in Liver (A), Spleen (B), Kidney (C), Gut (D), and Gill (E) of crucian carp ($n = 3$ fish/group) after immunization. The values are expressed as the means $\pm$ SD ($n = 3$). Different letters placed above represent significant differences between the treatment groups ($p < 0.05$).

accordance with the present results, crucian carp administration with different fermented cultures could enhance the average growth rate and specific growth rate. These findings indicated that probiotic fermented herbs could improve digestibility and enhance fish growth performance [33]. Hence, probiotic fermented herbs might have potential applications as feed additives. Bacterial antagonism demonstrated that the fermented cultures effectively antagonized *A. hydrophila*. This result further supported the idea of active components of fermented herbs could decrease pathogen growth, colonization and virulence [34].

Biofilm is a major factor causing long-term infection, virulence production and drug resistance of pathogenic bacteria [35]. Here, we demonstrated that fermentation cultures dose-dependent inhibition of *A. hydrophila* biofilm formation. Chen et al. reported that ethanolic

extract from DY could inhibit biofilm formation of methicillin-resistant *Staphylococcus aureus* (MRSA) [36]. Besides, *L. rhamnosus* was found that inhibit biofilm more strongly than *L. casei* [37]. In agreement with our data, *L. rhamnosus* fermented DY cultures could inhibit the biofilm formation of *A. hydrophila*. These results support that MIC and MBC determination, indicating that low concentrations of *L. rhamnosus* fermented DY cultures could effectively inhibit bacterial biofilm formation. It is possible to hypothesize that fermented cultures could affect the quorum sensing of bacteria by reducing the receptor activity of signaling molecules and ultimately regulating the biofilm. However, further research should be undertaken to investigate the inhibition mechanism of bacterial biofilm by fermentation cultures. Hydrolytic enzymes could be used as inducers to indirectly stimulate the active substances of

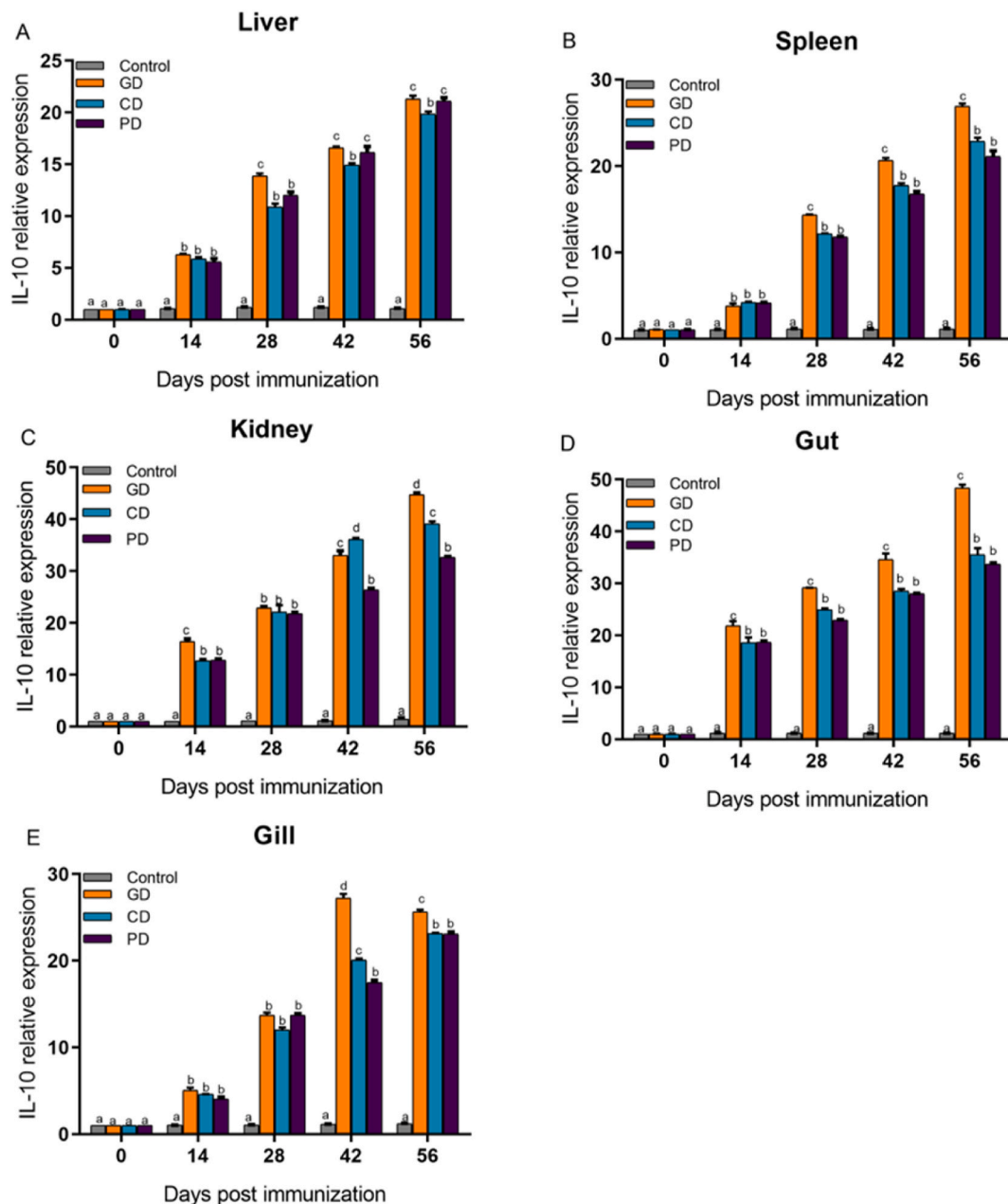


Fig. 9. qRT-PCR analysis of the expression of cytokine IL-10 gene in Liver (A), Spleen (B), Kidney (C), Gut (D), and Gill (E) of crucian carp ($n = 3$ fish/group) after immunization. The values are expressed as the means $\pm$ SD ($n = 3$). Different letters placed above represent significant differences between the treatment groups ($p < 0.05$).

herbal medicines and enhance the pharmacological effects [38]. We found that hydrolytic enzymes produced from gut microorganisms could be increased by oral probiotic fermentation of herbal medicines. We tested the activity of enzymes in fermentation cultures with agar plates and found positive reactions for protease, lecithin and urease. Zhang et al. [39] found that proteases may degrade macromolecules in the gastrointestinal tract of fish and facilitate the digestion and absorption of food. Hartmann et al. [40] fed lecithin-rich food to arthritic rats and the inflammation was alleviated. It might be hypothesized that fed probiotic fermented herbal cultures could promote fish growth and enhance animal immunity through a variety of hydrolytic enzymes.

Serum immune parameters are commonly used to analyze the innate immune defense, such as AKP, SOD, CAT, GSH-Px, LYS, IgM, complement C3 and C4 are important indicators of antioxidant capacity and resistance to pathogens in fish [41–43]. AKP is an extracellular enzyme

that regulates non-specific immunity and nutrient metabolism. Consistent with our results, prior studies have found that the AKP of Caspian white fish was significantly higher than the control group after dietary supplementation with *L. casei* ($p < 0.05$) [44]. SOD is a superoxide radical scavenger, which can work with CAT to break down superoxide radicals into H_2O and O_2 , and complete removal of excessive free radicals from fish [45]. In our experiments, we found that SOD and CAT activities were significantly increased after fish treated with fermented cultures. In addition, the activity of GSH-Px was also increased in the experiment. The elevated oxidative stress indicators predicted that probiotic fermented DY could protect fish tissues from oxidative damage. Lysozyme is an important defense protein of fish innate immunity, which can prevent pathogenic bacterial infection and activate the complement system [46]. Hossein et al. reported that the *Xiphophorus helleri* treated with *Lactobacillus acidophilus* could increase LYS and C3

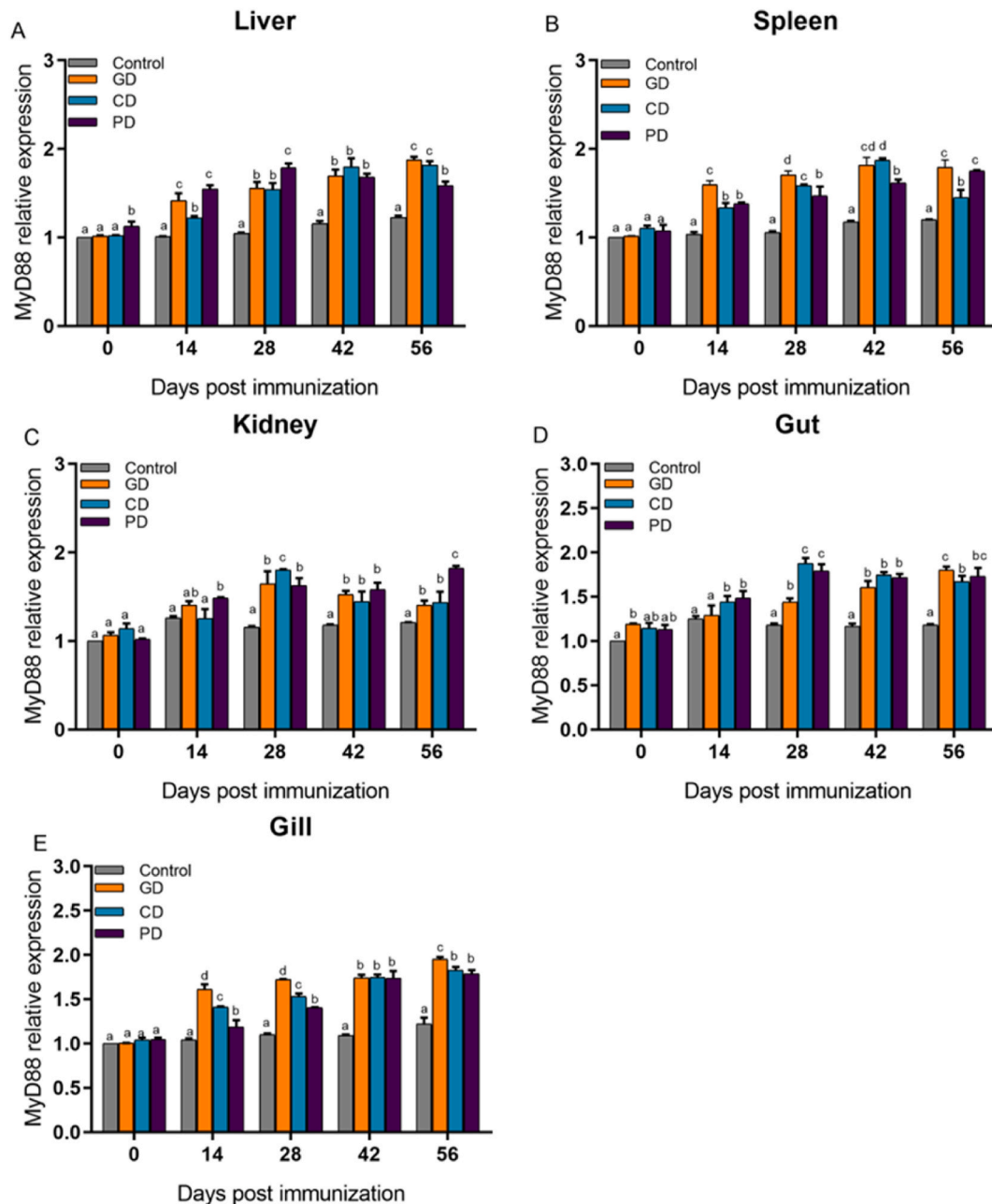


Fig. 10. qRT-PCR analysis of the expression of cytokine MyD88 gene in Liver (A), Spleen (B), Kidney (C), Gut (D), and Gill (E) of crucian carp ($n = 3$ fish/group) after immunization. The values are expressed as the means $\pm$ SD ($n = 3$). Different letters placed above represent significant differences between the treatment groups ($p < 0.05$).

levels [47]. Our research results suggested that probiotics fermented herbal medicine DY could promote the improvement of LYS, C3 and C4. The improvement of innate immune parameters could be contributed to the pathogen resistance in fish. IgM is an important immunoglobulin of the bony fish, which plays a key role in specific immunity. Our study found that probiotics fermented DY cultures could maintain high levels of IgM in fish. We hypothesized that probiotics fermented herbal cultures could reach the intestinal mucosa, and the components might penetrate into the lamina propria, thereby triggering B lymphocytes and inducing a local mucosal immune response. Subsequently, IgM was released into the bloodstream and stimulates a systemic immune response. Therefore, these results further supported the hypothesis that probiotics fermented with herbs could enhance the systemic response in bony fish.

Cytokines are proteins with functions related to tissue repair, cell

growth and immunomodulation. The pro-inflammatory cytokines *IL-1 β* and *TNF- α* could be triggered in response to bacterial and viral invasion [48]. *IL-1 β* can trigger the lymphocytes and macrophages and plays an important role in the early immune response [49]. Liu et al. reported that *IL-1 β* and *TNF- α* gene expression could be up-regulated through treating with *L. acidophilus* JCM 1132 in hybrid tilapia [50]. In this study, the expression level of *IL-1 β* in different tissues of fish received with fermented cultures was significantly higher than the control group. We speculated that fermented herbal cultures could promote pro-inflammatory cytokines in lymphocytes and macrophages. *IFN- γ* could enhance the resistance of animals to viral and bacterial invasion. In this study, fermented cultures promoted *IFN- γ* gene expression in crucian carp tissues. This result might be explained by the fact that activation of *IL-1 β* and *TNF- α* by macrophages could induce the production of *IFN- γ* [51]. In addition, we found that high expression of the

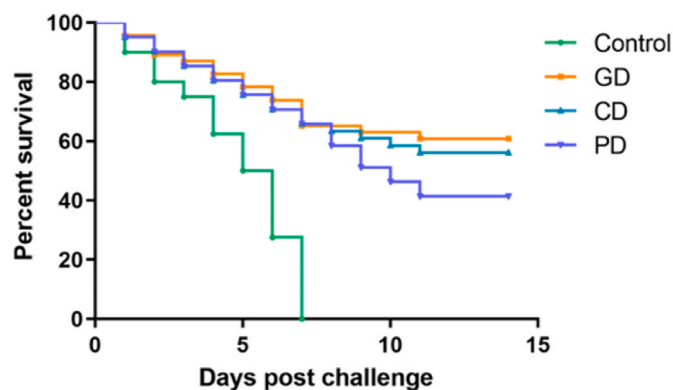


Fig. 11. The survival rate of fish immunized with lactic acid bacteria fermentation herbal DY culture and PBS respectively was examined by intraperitoneal injection of crucian carp with *A. hydrophila* after 56 days of immunization. 40 fish/group were used to record the survival rate for 14 days.

IL-10 gene in various tissues of crucian carp. Consistent with the present results, previous studies have shown that the active component in DY could stimulate higher *IL-10* gene expression in mice, zebrafish and other animals [52,53]. Kong et al. [54] reported that *Lactococcus lactis* L19 stimulated *IL-10* gene expression in snakehead fish (*Channa argus*). These results instruct us that the elevated *IL-10* gene expression may be due to synergistic effect of the combination of herbal medicine and

probiotics. We assumed that *IL-10* levels were elevated by fermented cultures stimulated when inflammation was triggered by tissue damage with fish. *IL-10* gene might down-regulate the monocyte surface major histocompatibility complex II (MHC II) expression, and reduce its antigen-presenting effect. Moreover, T lymphocyte activity was decreased, thereby inhibiting the activation, migration and adhesion of inflammatory cells. Finally, these cascade responses exhibited anti-inflammatory effects and protected fish from tissue damage. We found that *MyD88* levels were elevated in crucian carp tissues. The results might be explained by the fact that the fermented cultures may be recognized by TLR4 receptors in the Toll receptor family and delivered to *MyD88*, further activating the NF- κ B signaling pathway to induce immune response.

A. hydrophila is the main pathogenic bacteria that cause bacterial septicemia and inflammation in fish and seriously threatens the aquaculture industry [55]. Our results showed that all three lactic acid bacteria fermented herbal DY cultures could protect fish against *A. hydrophila* infection. Compared to the control group, the crucian carp received with fermented cultures showed a higher survival rate. Giri [56] and Yang [57] et al. also verified that both dietary supplementations with *L. plantarum* and the herbal medicine salidroside could protect fish from *A. hydrophila* infection. Interestingly, *L. rhamnosus* fermented herbal DY had the strongest protective effect. The high protection rate may be due to the fact that probiotic fermented herbal cultures could activate the innate immune response in fish and further promote the local mucosal immune response. Cytokines and antibodies were secreted to produce a protective effect against pathogenic bacteria.

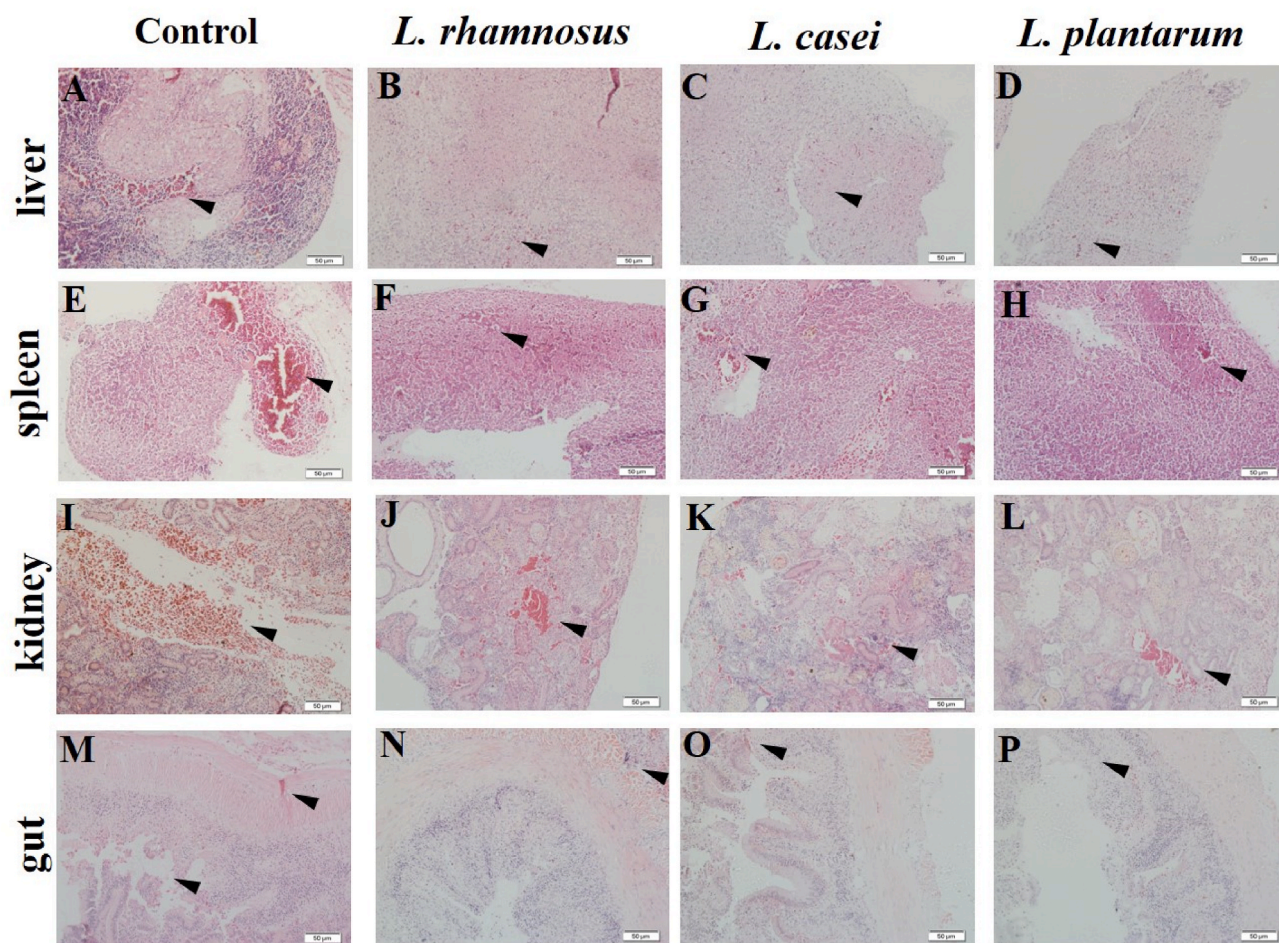


Fig. 12. Histopathological sections of liver (A, B, C and D), spleen (E, F, G and H), kidney (I, J, K and I) and intestine (M, N, O and P) of crucian carp fed fermented herbal DY cultures after challenge. The black arrow indicates the site of hemorrhagic lesions in crucian carp after challenge with *A. hydrophila*. Data are representative of three independent experiments. (Scale bars, 50 μ m).

A. hydrophila is widely present in aquatic products and aquaculture environments and could cause severe septicemia, diarrhea and enteritis in fish [58]. We also observed severe hemorrhage in various tissues of the tract by pathological histological analysis. Furthermore, fermented cultures were the most effective in treating intestinal damage in crucian carp, compared to other tissues. This could be due to the destruction of small intestinal villi and crypt after infection of crucian carp by *A. hydrophila*. But the probiotic fermented cultures, which reduced the damage of intestinal structure by pathogenic bacteria, promoted the secretion of mucin by goblet cells and increased the intestinal tight junction protein content to protect the intestinal health.

However, further research trials and analyses are needed to separate herbal and probiotic bacteria to modulate the immune response and disease resistance in fish. Future studies should focus on the compositional changes of herbal medicines after being fermented by probiotics, and the immunomodulatory mechanisms involved in specific components. This will provide an important research base for the development of new herbal microecological formulations.

5. Conclusion

Scientific evidence support that probiotic fermented herbal products could improve disease resistance and health in fish is scarce. However, this study provides evidence for understanding inflammation to *A. hydrophila* infections by which probiotic fermented *Sanguisorba officinalis* in fish could regulate the innate immune response and disease resistance. Furthermore, our findings emphasize the importance of probiotic fermentation for immunomodulatory and growth performance. This finding has important implications for developing alternative strategies to replace antibiotics, and probiotic fermented herbal cultures-containing dietary could be used for safe *A. hydrophila* decolonization strategy. Such a probiotic fermentation approach would have numerous benefits over the typical treatment involving antibiotics, which is aimed effectively at preventing the pathogen infection.

CRediT authorship contribution statement

Tao Wang: performed the experiment and prepared the manuscript. **Xin-Lei Tian:** contributed to bacteria cultivation. **Xue-Bin Xu:** was involved in experiment design and manuscript revision. **Hui Li:** involved in manuscript revision. **Ye Tian:** involved in manuscript revision. **Yi-Han Ma:** involved in manuscript revision. **Xiao-Fei Li:** involved in manuscript revision. **Na Li:** involved in manuscript revision. **Ting-Ting Zhang:** involved in manuscript revision. **Yu-Di Sheng:** involved in manuscript revision. **Qian-Xi Tang:** was involved in manuscript revision. **Lei Zhang:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Chun-Feng Wang:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Shahrood Ahmed Siddiqui:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Li-Xia Wang:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Xiao-Feng Shan:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Ai-Dong Qian:** contributed to interpretation of data, and the critical revision of the study methods, Data curation. **Dong-Xing Zhang:** conceived the idea and designed the experiment, All the authors have read and approved the final manuscript.

Declaration of competing interest

The authors declare that they have no conflict of interest.

Data availability

Data will be made available on request.

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